

Sreeja Reddy Yeluru

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EDUCATION

Master of Science, Artificial Intelligence, DePaul University, Chicago, IL Jun 2026
GPA 4.0, Presidential Scholarship
Coursework: Artificial Intelligence I and II, Natural Language Processing, Neural Networks and Deep Learning, Programming Machine Learning Applications, Advanced Machine Learning, Image Processing, Applied Algorithms and Structures

Bachelor of Engineering, Computer Science, MVJ College of Engineering, Bangalore, India Aug 2021
CGPA 8.71

EXPERIENCE

Research Assistant, DePaul University Sep 2025 - Present | Chicago, IL

- Implementing the harmonic persistence barcode, a new topological descriptor for machine learning, translating mathematical specifications into Python with NumPy, SciPy, and GUDHI.
- Built the pipeline from point clouds to barcodes, including boundary and chain matrices, Gram-Schmidt orthogonalization, and generator cycle extraction, with correctness checks.
- Designing controlled experiments against traditional persistence barcodes using persistence images and SVM classifiers on geometric shapes and generated point clouds.

Application Development Associate to Analyst, Accenture Oct 2021 - Aug 2024 | Bangalore, India

- Supported the core trading application of a global oil trading client, a system at the center of its trading operations, working with many upstream and downstream teams.
- Diagnosed a client-reported logic issue on a trading screen by tracing its SQL queries, and the fix was deployed that weekend.
- Reduced recurring support tickets by 20% by finding the most frequent issues and delivering a self-service fix that resolved them without reaching the support team.
- Monitored database health and usage, investigated unusual load and CPU spikes to prevent repeats, and supported DB2 upgrades, data migrations, and major data backups.
- Supported daily market price loads, period-end P&L reconciliation, year-end processing, and monthly production releases, and was on the front line during major outages.

Web Developer Intern, ClearExam Nov 2020 - Jan 2021 | Remote

- Built responsive, cross-device pages for an ed-tech platform from design mockups using HTML, CSS, and JavaScript.

Data Science Intern, Verzeo Aug 2020 - Nov 2020 | Remote

- Built a Random Forest churn prediction model with 85% accuracy after exploratory data analysis and feature engineering, and reported at-risk accounts in a Power BI dashboard.

PROJECTS

WanderWear: Multi-Agent Travel Planner and Personal Stylist | [Live](#) | [Overview](#)

- Designed, built, and deployed a production multi-agent LLM application on LangGraph, taking it from hand-drawn wireframes to a live product used through a web and mobile PWA.
- Built agentic workflows in which a travel agent calls live weather, maps, and routing tools and hands off to a vision-based stylist agent, with an MCP server exposing five tools.
- Reduced hallucinations with a critic model scoring each plan against tool results, database validation of every returned item, and token and step budgets as guardrails.
- Built LLM evaluation and observability: an LLM-as-judge eval gate that blocks regressions and per-run logging of cost, latency, tool calls, and quality score.
- Scaled to 10,000 concurrent plan submissions with a bounded job queue on a stateless FastAPI service, and cut worst-case latency from 484 s to 62 s with a fallback provider and circuit breaker.

- Secured multi-user data with PostgreSQL row-level security and per-request authentication, verified by isolation tests and a live security audit that fixed five findings.
- Deployed on AWS (ECS Fargate, ECR, API Gateway, Amplify, Secrets Manager, CloudWatch) with CI/CD through GitHub Actions, OIDC, and Docker, running 741 automated tests per release, and cut hosting cost about 60% through an API Gateway design.

Agentic RAG for CleanTech Question Answering

- Built a multi-tool LangChain agent over 149,504 document chunks in ChromaDB with retrieval, summarization, Semantic Scholar paper search, and reasoning tools.
- Improved LLM-as-judge answer quality from 3.9 to 4.3 out of 5 over a baseline RAG pipeline on 50 questions, measured with BERTScore (F1 0.85), ROUGE-L, and retrieval nDCG (0.67).

AI Policy RAG | [Live](#)

- Built a RAG application with cited answers over four AI policy documents, running Qwen2.5-1.5B locally with LangChain, Chroma, and sentence-transformer embeddings.
- Compared three embedding models and tuned chunk size, overlap, and top-k for retrieval quality.

Malicious Prompt Detector | [Live](#)

- Detected prompt injection and jailbreak attempts with 95% accuracy, 0.95 F1, and 0.98 ROC-AUC on 39,000 prompts using a stacking ensemble (Logistic Regression, LinearSVC, LightGBM) on TF-IDF features.
- Reduced overfitting gaps by up to 51% through hyperparameter tuning, 5-fold cross-validation, and bias-variance analysis.

Energy Load Forecasting

- Reached 1.4% MAPE on hourly electricity demand with a CNN-LSTM in PyTorch, against 3.37% for an LSTM and 13.42% for an RNN, on 66,000 hourly readings merged with weather data (team of four).

Reinforcement Learning and RLHF

- Implemented supervised fine-tuning, reward model training, and PPO for RLHF on GPT-2 with Hugging Face Transformers and TRL.
- Continued training a Deep Q-Network for Atari Pong, improving average reward by 34%, and tuned DQN, PPO, and A2C across three autonomous driving environments with Stable-Baselines3.

SKILLS

Languages: Python, SQL, JavaScript, Java, COBOL

Generative AI and LLMs: AI agents, agentic workflows, multi-agent systems, tool calling, LangGraph, LangChain, Model Context Protocol (MCP), RAG, embeddings, vector databases (ChromaDB), prompt engineering, LLM evaluation (LLM-as-judge), guardrails, hallucination mitigation, fine-tuning, RLHF, Hugging Face Transformers, OpenAI, OpenRouter, and Groq APIs

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, LightGBM, deep learning, CNN, LSTM, ensemble methods, NLP, computer vision (OpenCV), reinforcement learning, time-series forecasting, model evaluation, hyperparameter tuning, cross-validation

Data Science: pandas, NumPy, SciPy, exploratory data analysis, feature engineering, data visualization, Power BI, experiment design

Software and Cloud: FastAPI, REST APIs, async Python, PostgreSQL, Supabase, Next.js, React, Docker, AWS (ECS Fargate, ECR, API Gateway, Amplify, IAM, Secrets Manager, CloudWatch), GitHub Actions, CI/CD, pytest, Git, system design

Research: topological data analysis, persistent homology, GUDHI, linear algebra, scientific computing

PUBLICATION

House Price Prediction Using Machine Learning. International Journal for Research in Applied Science and Engineering Technology (IJRASET), Volume 9, Issue 8, August 2021. Co-author.

CERTIFICATIONS

Programming Languages (C, C++, Java, Python, MySQL), IIT Bombay
Front-End Web Development, Cambridge Certification Authority